

INLA fit to Poisson vs GAM

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Data cleaning

- afs or marriage age greater than age
- married but no afs

1 Age at marriage (AAM) model

Five possible scenarios outlined in Figure 1 where time *at risk of getting married* is highlighted.



Figure 1: AAM risk set model

The first case is the typical states transition with observed marriage event; the second case is right censored at the time of interview. The third and fourth case are special in that the time “at risk of getting marriage” are counted from birth. To reflect the difference between this exposure time and the two standard cases (1 and 2), respondent’s age at the exposure time are taken into account.

1.1 Time-split data format

The exposure time is divided by equidistant intervals of length $\tau = 1$, from zero year to maximum 46 in this data. Status of the respondents at all periods prior to the end of the study period (Figure 1) was set to single.

Respondent’s age at each of the intervals is calculated accordingly and modelled as a covariate. To avoid dropping of samples with immediate transition (time at risk is zero), a month was added to the age at marriage if age at marriage equal age at first sex (or adding 1/12 to exposure time where exposure time is zero). The example is shown in Table 1.

```
lexis %<>% mutate(time = if_else(time == 0, time + 1 / 12, time)) # add a
month

## NOTE: entry.status has been set to "Single" for all.
## NOTE: entry is assumed to be 0 on the tar timescale.
## NOTE: Dropping 5912 rows with duration of follow up < tol
```

Table 1: Lexis data format: lex.id: respondent’s id, tar: time at risk, lex.dur: at risk duration, aar: age at risk, age: age at interview,

lex.id	tar	lex.dur	lex.Cst	lex.Xst	age	marriage_age	afs	event	age_0	time	aar
2	0	1	Single	Married	31	15	14	1	14	1	14

lex.id	tar	lex.dur	lex.Cst	lex.Xst	age	marriage_age	afs	event	age_0	time	aar
4	0	1	Single	Single	31	14	16	1	0	14	0
4	1	1	Single	Single	31	14	16	1	0	14	1
4	2	1	Single	Single	31	14	16	1	0	14	2
4	3	1	Single	Single	31	14	16	1	0	14	3
4	4	1	Single	Single	31	14	16	1	0	14	4

Transitions: To From Single Married Records: Events: Risk time: Persons:
Single 133958 12837 146795 12837 146795 17980

1.2 Poisson model - counts data format

Counting the events and person-year by time at risk and age at risk (Table 2).
The empirical rates of marriage are shown in Figure 2 where time since at risk
is grouped for visualization.

`## summarise()` has grouped output by 'tar'. You can override using the ``.groups`` argument.

Table 2: Count data format

tar	aar	count	person_year
0	0	0	6272
0	8	0	91
0	9	0	20
0	10	7	101
0	11	13	138
0	12	61	451

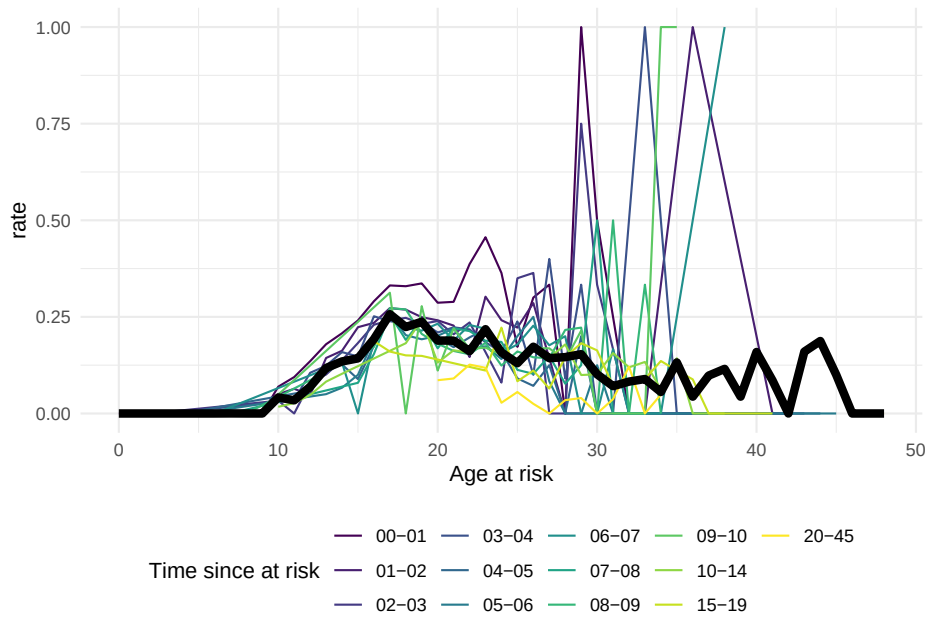


Figure 2: Rate of marriage per person-year by exposure time and age at risk based on count format data. Thick line is median across exposure time.

1.3 Smoothed age effect and tar - GAM model

To reflect the observed nonlinear age effect in Figure 2. Age effect is smoothed with `gam` as follows:

The model with smoothed age effect shows significant improvement, as seen in in Figure 4. Rate of marriage increases with age and tapering off from age 25. Consider adolescent age, marriage are more likely to occur within a year exposed to risk (sexual debut in this case).

`## Warning: Removed 17 rows containing missing values (geom_point).`

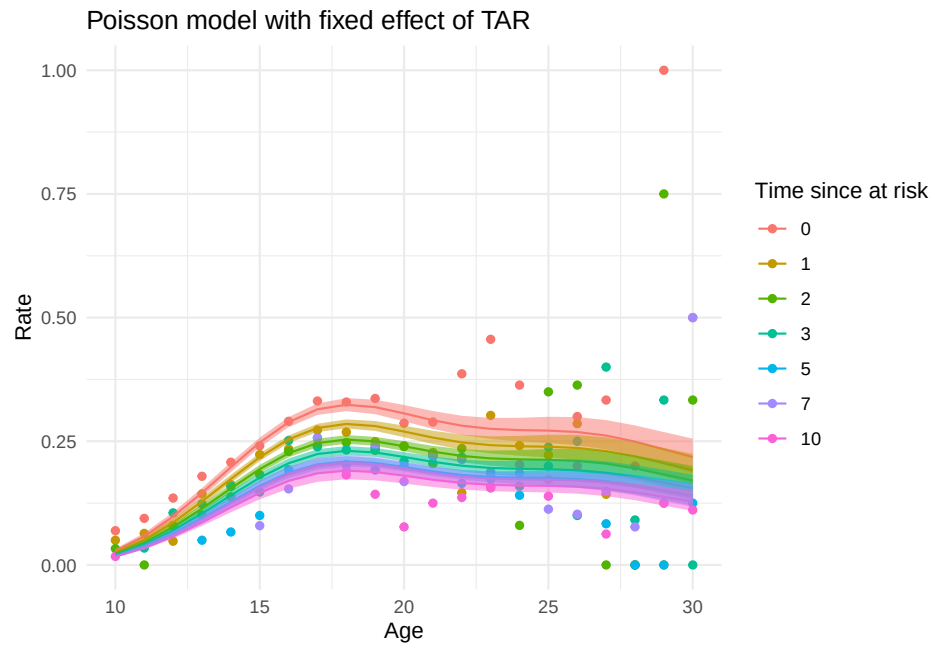


Figure 3: Estimate rate of marriage by age

Warning: Removed 17 rows containing missing values (geom_point).

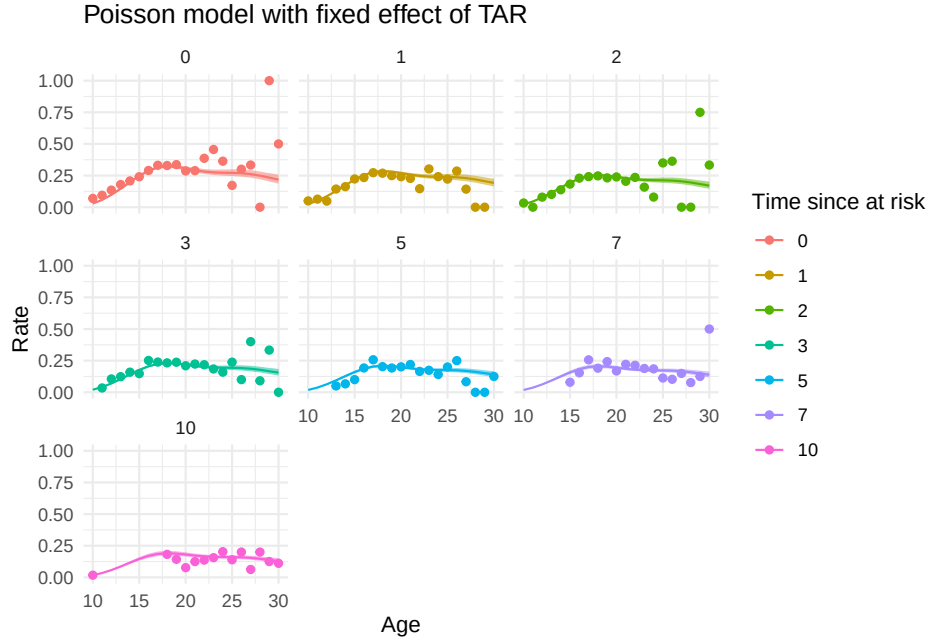


Figure 4: Estimate rate of marriage by age

1.4 Poisson vs Zero-inflated poisson with INLA

The likelihood is defined as

$$\text{Prob}(y \mid \dots) = p \times 1_{[y=0]} + (1 - p) \times \text{Poisson}(y)$$

where p is a hyperparameter where

$$p = \frac{\exp(\theta)}{1 + \exp(\theta)}$$

and the Gaussian prior is defined on θ .

Table 3: WAIC Poisson vs Zero-inflated Poisson

model	waic	p.eff
Poisson	1792.911	41.26568
Zero-inflated	1799.138	38.61314

Estimate the probability of having no case of marriage is 7%. This is not what

we want. The exceed zeros in is the zero in time, not zero in number of marriages in a period.

Let work on this to get INLA into the story. Mainly how to deal with prediction without the offset: fitted values will include `offset` but does not include `E`

Table 4: Poisson zero-inflated hyperparameters

	mean	sd
zero-probability parameter for zero-inflated poisson_1	0.0079547	0.005240
Precision for aar	3.5270399	1.067209
Precision for tar	81.5300983	33.072686

Table 5: Poisson hyperparameters

	mean	sd
Precision for aar	2.647194	0.8136148
Precision for tar	77.444881	31.6170538

Effect of AAR and TAR are shown in Figure 5.

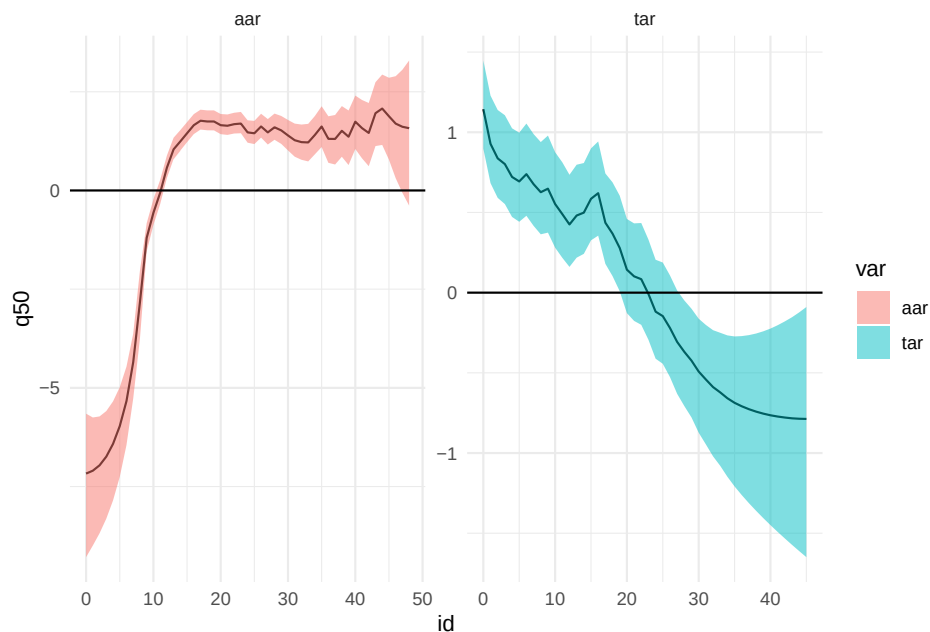


Figure 5: RW2 models for age and time at risk

```
## Warning: Removed 147 rows containing missing values (geom_point).
```

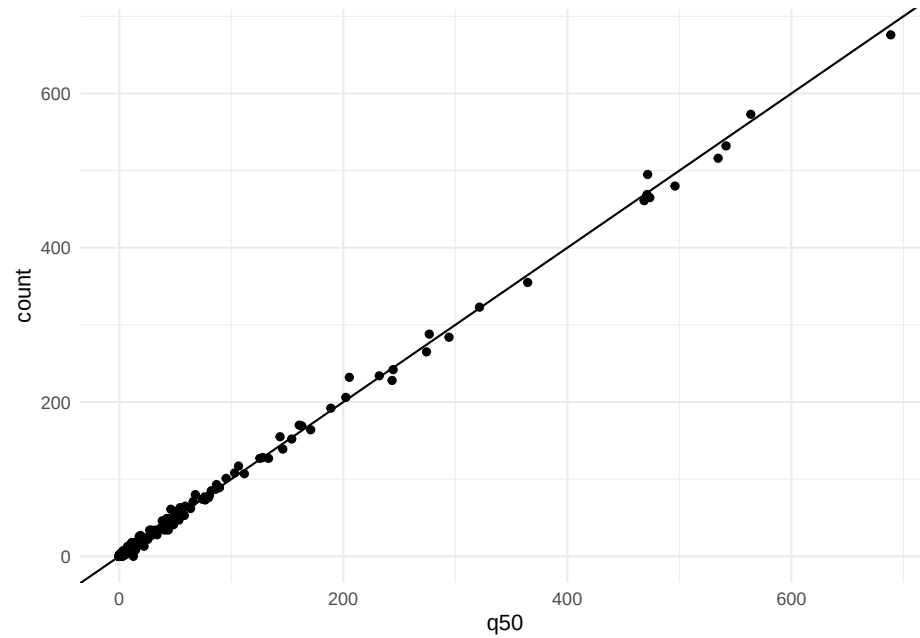
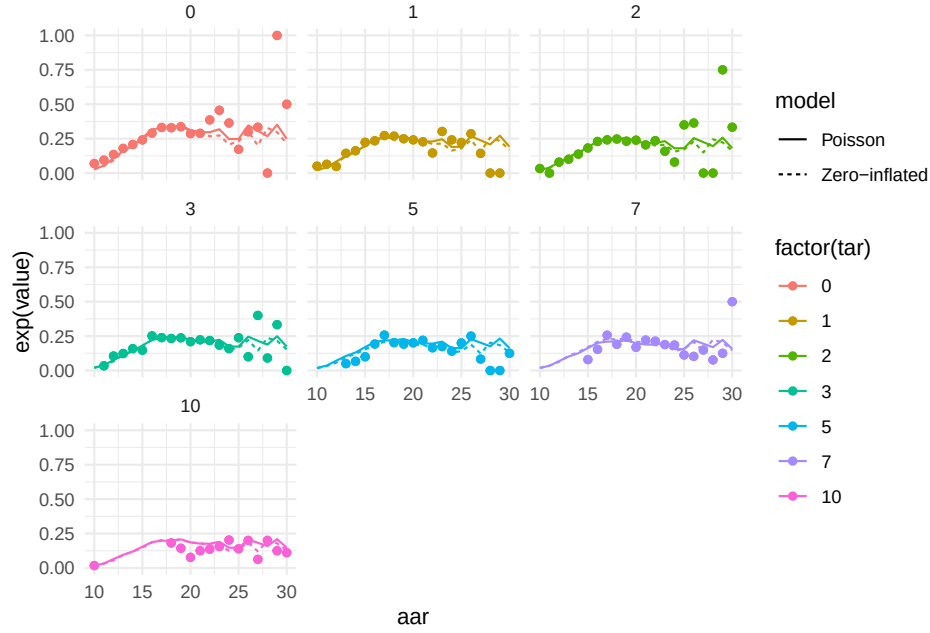


Figure 6: Fitted values vs median - INLA

Sample from posteriors and get the prediction for the combinations of TAR and AAR.

```
## Warning: Removed 34 rows containing missing values (geom_point).
```



References

A Appendix

A.1 Splitting time - collecting...

The common survival analysis views time as the response variable. If survival model is viewed from the demographic aspect, the basic observation is not one time to event (or censoring) for each individual, but rather many periods of follow up from each individual. This means modeling of rates rather than time to response, where response is a dichotomous outcome in each interval.

In this set up, time is a covariate and exposure time is response. Stratified analysis becomes an interaction between time and a categorical covariate, and time-varying coefficients becomes interactions between time and a continuous covariate. Finally, the modeling can be done with Poisson regression framework.

The Cox-model is a special case of a Poisson model where the time covariate is modelled with one parameter per failure time. Poisson modeling of disease rates and follow-up studies are usually restricted to constant rate over time-spans, If follow-up time are in small intervals, smoothing of rates can be done by splitting the follow-up in small intervals, and modeling datasets in which each record represents a period of follow-up time for a person.

Takes the survival X as response variable and censoring by time Z . In survival data we only observe the time $\min(X, Z)$ and the event indicator $\delta = 1_{X < Z}$.

In a life-table, differences on the time scale are accumulated as risk time whereas the position on the time scale (age) for these are used as a covariate.

Consider a study where the follow-up time for each individual is divided into small intervals of equal length y and each with an exit status recorded (0 for most of the intervals and 1 for the last interval for individuals experiencing an event). Each interval contributes an observation of an empirical rate, (d, y) , where d is the number of events in the interval, and y is the length of the interval, i.e. the risk time. This definition is slightly different from the traditional as d/y (or $\sum d / \sum y$) in that it keep the entire information content in the observation even if the number of events is 0. This makes it usable as a response variable in all situations.

The rate of event occurrence is defined as a function of some timescale, t :

$$\lambda(t) = \lim_{h \rightarrow 0} \frac{P(\text{event in } (t, t+h] | \text{at risk at time } t)}{h}$$

The rate may depend on covariates. In this formulation time t is a covariate and h is risk time, namely the difference between two points on the time ($t+h$ and t).

The likelihood contribution from an observed (d, y) for an interval with constant rate λ is a Bernoulli likelihood with probability λy

$$L(\lambda | (d, y)) = (\lambda y)^d (1 - \lambda y)^{1-d} = \left(\frac{\lambda y}{1 - \lambda y} \right)^d (1 - \lambda y)$$

$$\log(L) = l(\lambda | (d, y)) = d \log\left(\frac{\lambda y}{1 - \lambda y}\right) + \log(1 - \lambda y)$$

The contributions to the likelihood from one individual are conditionally independent, i.e., the total likelihood from one individual across intervals t_0, t_1, t_2, t_3, t_4 is

$$P(\text{event att}_4 | \text{alive att}_0) = P(\text{event att}_4 | \text{alive att}_3) \tag{1}$$

$$\times P(\text{survive}(t_2, t_3) | \text{alive att}_2) \tag{2}$$

$$\times P(\text{survive}(t_1, t_2) | \text{alive att}_1) \tag{3}$$

$$\times P(\text{survive}(t_0, t_1) | \text{alive att}_0) \tag{4}$$

which is similar to independent Poisson observations likelihood. The amount and spacing of events limits how detailed the rates can be modelled. The model depends on how large intervals of constant rate is acceptable to the study context.

The Cox model specifies the intensity λ as a function of time (t) and the covariates $(x_1, \dots, x_p)$ through the linear predictor $\eta_i = \beta_1 x_{1i} + \beta_p x_{pi}$ as:

$$\lambda(t, x_i) = \lambda_0(t) \exp(\eta_i)$$

with the partial log-likelihood

$$l(\beta) = \sum_{\text{death times}} \log\left(\frac{e^{\eta_{death}} \sum_{i \in R_t} e^{\eta_i}}{\sum_{i \in R_t} e^{\eta_i}}\right)$$

where R_t is the risk set at time t including all individuals at risk at time t

The assumption behind the Poisson approach is essentially only the assumption that a model with constant rates in each small interval gives an adequate description of data. So in practice we would split data in small equidistant intervals